P101 · Unit 5

Circular motion and gravitation

Recall: retrieve readily · Understand: reconstruct the reasoning · Consult: look up as needed. Conditions are part of each result.

Use	Topic	Result and conditions
Recall	**Timing and speed**	$f = 1 / T$; $v = 2\pi r / T = 2\pi r f$. Uniform circle: r radius (m), T period (s), f frequency (Hz), v speed (m/s). Lesson 1
Recall	**Angular measure**	$\theta = s / r$ in radians for arc length s. Uniform angular speed $\omega = 2\pi / T$ (rad/s); $v = r\omega$ and $a_r = r\omega^2$. A full turn is 2π radians. Lesson 1
Understand	**Vector change**	$\Delta v = v_2 - v_1$; $\overline{a} = \Delta v / \Delta t$. Subtract vectors, not speeds. A finite average need not equal an instantaneous acceleration. Lesson 1
Recall	**Radial acceleration**	$a_r = v^2 / r = 4\pi^2 r / T^2$, inward. Velocity is tangent. For uniform speed, acceleration is entirely radial; if speed varies, a tangential component also exists. Lesson 1
Understand	**Force inventory**	$\sum F_{\text{inward}} = mv^2 / r$ for the radial component of Newton's second law. Draw real interactions only. "Centripetal" names the inward resultant, not an extra force. Lesson 2
Consult	**Friction and support**	Flat horizontal circle, weight/contact only vertically: $N = mg$. Sticking requires $mv^2 / r \le \mu_s N$; hence $v_{\max} = \sqrt{\mu_s g r}$. μ_s is a dimensionless friction coefficient. Lesson 2
Consult	**Conical string**	String length L, angle θ to vertical: $r = L\sin\theta$; $S\cos\theta = mg$; $S\sin\theta = mv^2 / r$. Thus $v^2 = rg\tan\theta$ for $0 < \theta < 90^\circ$. S is tension. Lesson 2
Understand	**Vertical circle**	Top: $S + mg = mv_t^2 / r$. Bottom: $S - mg = mv_b^2 / r$. Inward positive at each. A cord needs $S \ge 0$; top condition $v_t \ge \sqrt{gr}$ is local, not a complete launch criterion. Lesson 2
Understand	**Physical feasibility**	An unattached contact needs $N \ge 0$; a cord cannot push. Negative required support means the assumed constrained path fails. Do not take its absolute value. Lesson 2

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Gravity and circular models

Use	Topic	Result and conditions
Recall	**Gravity and field**	$F = GMm / r^2$; $g(r) = GM / r^2 = \mu / r^2$; $\mu = GM$. Attractive pair forces have equal magnitudes on different recipients. Exact for point masses or non-overlapping spherical mass distributions. Lesson 3
Understand	**Distance and units**	$r = R + h$: centre distance, source radius and altitude. $G \approx 6.67430 \times 10^{-11}$ N m²/kg²; μ in m³/s²; g in m/s² or N/kg. Weight mg differs from a support reading. Lesson 3
Recall	**Circular orbit**	$v = \sqrt{\mu / r}$; $T = 2\pi\sqrt{r^3 / \mu}$. Exterior circular path, dominant spherical source treated as fixed, gravity only. Velocity must be tangential. Satellite mass cancels. Lesson 3
Understand	**Controlled ratios**	$k = r / r_0$, $q = \mu / \mu_0$: $v / v_0 = \sqrt{q / k}$; $T / T_0 = \sqrt{k^3 / q}$; $g / g_0 = q / k^2$. Each point is a separate scenario, not time evolution. Lesson 4

Consult · Python lookup and checks

```
import numpy as np
# Finite SI values: mu > 0, r > source radius > 0.
v = np.sqrt(mu / r)
T = 2 * np.pi * r / v
g = mu / r**2
```

Assign inputs before these formula lines. For arrays and input validation, use the Lesson 4 function. Hand benchmark: μ = 8 × 10¹² m³/s², r = 2 × 10⁶ m, source radius 10⁶ m → v = 2000 m/s, T = 2000π s, g = 2 m/s². Check units, exterior domain, benchmark and ratios before interpreting plots. Numerical consistency is not observational validation.

Return to the unit teaching and exercises for derivations, examples and feedback.

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